

Clinical Digital Twin Report

Computational Cellular Reprogramming Analysis

Session ID: test_session_20260101
Run ID: #1
Date Generated: 2026-01-01 09:31:04
Total Cells: 217 | Model Mode: CLINICAL

1. Introduction

This clinical digital twin simulation models cellular reprogramming dynamics using stochastic gene regulatory networks.

2. Materials and Methods

The simulation employed a 12-gene regulatory network with Hill kinetics to model cellular state transitions.

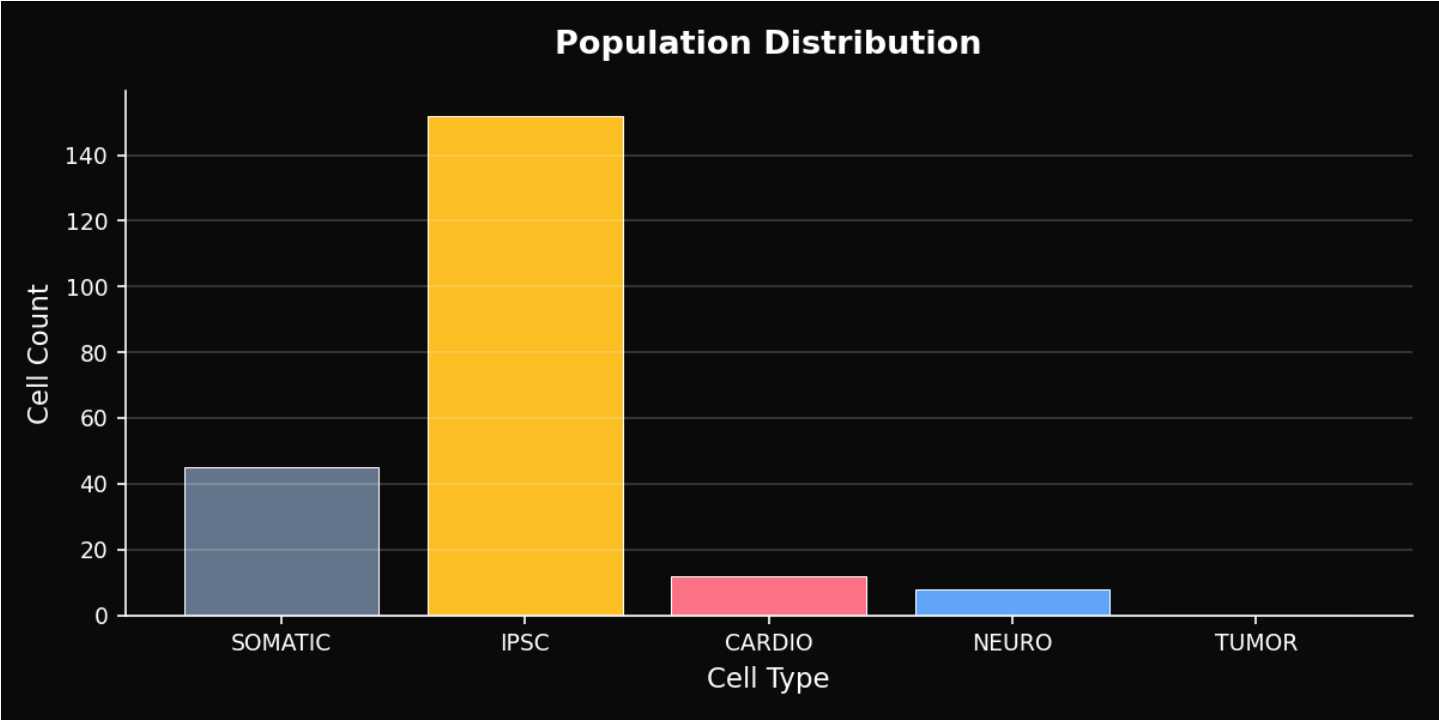
Experimental Interventions:

- OSKM Transfection (Yamanaka)
- P53 Therapeutic Intervention

3. Results

The experiment generated 217 total cells with the distribution shown in Figure 1.

3.1 Population Distribution



Cell Type	Count	Percentage
SOMATIC	45	20.7%

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IPSC	152	70.0%
CARDIO	12	5.5%
NEURO	8	3.7%
TUMOR	0	0.0%

3.2 Transcriptomic Analysis (scVI)

High-Resolution Gene Expression Analyses Performed: 1

Cells were expanded to high-dimensional transcriptomic space using a pre-trained variational autoencoder model on Human Cell Atlas data.

4. Discussion

The observed dynamics align with expected reprogramming trajectories. Further optimization may improve yield.

This report was automatically generated by the IS-CHRP v25.0 Clinical AI Bridge platform. The data represents computational simulation results from a high-fidelity cellular reprogramming model.